

Action Horizon Ablation for Imitation Learning in Robot Manipulation

Motivation

In imitation learning for robot manipulation, a policy can be trained to predict either a single action per observation (single-step) or a short sequence of future actions (action chunking) from one observation. Action chunking has been proposed in recent work (e.g. ACT, Diffusion Policy) as a way to produce smoother, more temporally consistent behavior and to reduce compounding error from noisy single-step predictions. However, the right chunk length is not obvious: too short and the policy doesn't benefit from temporal consistency; too long and the policy commits to open-loop behavior that can drift before re-observing the environment. This study empirically measures how action horizon length affects task success rate on a simulated pick-and-lift task.

Setup

- **Task:** RoboSuite **Lift** (Panda arm, single rigid cube)
- **Demonstrations:** 150 expert demos, replayed from the official RoboSuite/robomimic Lift dataset through the simulator to generate full state observations
- **Observation space:** 14-dim state vector — end-effector position (3), end-effector orientation as quaternion (4), joint positions (7). No visual input.
- **Policy architecture:** 3-layer MLP (hidden dim 256, ReLU activations, tanh-bounded output), trained via behavior cloning (MSE loss between predicted and expert actions)
- **Action representation:** Native joint-space actions (7-dim: 6 joint deltas + gripper), held constant across all conditions
- **Variable:** Action horizon — the number of future timesteps predicted per forward pass, executed open-loop before re-observing
- **Evaluation:** 50 held-out episodes per condition; success defined as cube height exceeding the lift threshold (reward > 0.9 under RoboSuite's shaped reward)

Results

Horizon	Success Rate
1	22%
8	32%

16	62%
32	56%

Discussion

Success rate increased substantially as horizon grew from 1 to 16 — nearly tripling from the single-step baseline. This supports the hypothesis that short-horizon chunking helps the policy produce smoother, more temporally coherent motion, reducing the jitter and inconsistency that comes from re-predicting independently at every timestep.

Performance did not continue to improve monotonically, however. At horizon 32, success rate dropped to 56%, below the horizon-16 peak. This is consistent with the known tradeoff in open-loop action chunking: longer horizons mean the policy commits to more steps before it next observes the true state of the environment, so small early errors in the predicted sequence have more opportunity to compound before any correction occurs. The result suggests an inverted-U relationship between horizon length and performance, with the optimum in this setting falling around horizon 16 — short enough to limit compounding drift, long enough to gain the smoothing benefit of chunked prediction.

Limitations

- Single task (Lift) and single seed per condition; results may not generalize to more complex manipulation tasks or different demonstration distributions
- A simple MLP architecture was used in place of more expressive sequence models (e.g. diffusion-based action heads), which may respond differently to horizon length
- 150 demonstrations is a relatively small dataset; results may shift with significantly more data
- Demonstrations were derived by replaying recorded action sequences through the simulator rather than collected live, which may introduce minor distributional differences from native teleoperated data

Conclusion

Action chunking provides a substantial and non-trivial benefit over single-step prediction in this imitation learning setting, but the benefit is not unbounded — there is a clear optimal horizon length beyond which performance degrades due to compounding open-loop error. Horizon selection should be treated as a tunable hyperparameter rather than "longer is always better."